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Math 400

Math 400 Exam 1 Part 4 Due by Monday, September 21 11:59 pm

Instructions: Please show an appropriate amount of work for each problem. Write legibly and clearly. Please circle your answer. Make sure to simplify all answers.

You can print out this exam and write your solutions directly on the exam. If you are unable to print out the exam, then you can write your solutions on separate pieces of paper. If you write your solutions on separate paper, then **make sure to put the problems in numerical order**. **Please write your name on the exam.**

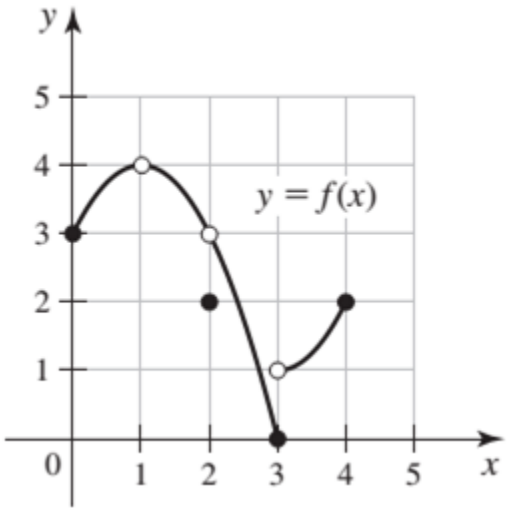
You'll have to submit Part 4 of the exam to me in **Canvas**. Please scan your exam and upload your **single** PDF file to Canvas. **Look under Assignments – Exams – Exam 1 Part 4.**

****This is an exam and so you shouldn't be working on the exam with other people and/or use apps/websites that can solve the problem for you.**

Name: _____

Math 400 Exam 1 Part 4 30 Total Points

(7 pts) 1. You are given the graph of a function f. Find the following limits, if they exist. If the limit does not exist, then explain why using **appropriate limit notation**.



$(a.)$ $\lim_{x\to 2} f(x)$

$(b.)$ $\lim_{x\to 3} f(x)$

$(c.)$ $\lim_{x\to 0^+} f(x)$

(6 pts) 2. Sketch the graph of an example of a function f that satisfies all of the given conditions. Include dashed lines for any asymptotes. Include the equations of any asymptotes. Answers may vary.

$f(0) = 5$ $\lim_{x \to 0^-} f(x) = 5$ $\lim_{x \to 0^+} f(x) = 1$

$\lim_{x \to -\infty} f(x) = 0$ $\lim_{x \to -3^-} f(x) = \infty$ $\lim_{x \to -3^+} f(x) = -\infty$

$\lim_{x \to \infty} f(x) = 4$

(7 pts) 3. Find the following limit **algebraically**. I want the exact limit, not a decimal approximation.

$$\lim_{x\to\infty}\left(11x-\sqrt{121x^2+2x}\right)$$

2 Precise Definition of a Limit Let f be a function defined on some open interval that contains the number a, except possibly at a itself. Then we say that the **limit of $f(x)$ as x approaches a is L**, and we write

$$\lim_{x \to a} f(x) = L$$

if for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that

$$\text{if} \quad 0 < |x - a| < \delta \qquad \text{then} \qquad |f(x) - L| < \varepsilon$$

(5 pts) 4. Prove the statement by using the **Precise Definition of a limit (that is, the ε, δ definition of a limit.)**

$$\lim_{x \to 7}(10x + 3) = 73$$

(5 pts) 5. Find the following limit **analytically**. If the limit is ∞ or $-\infty$, then state this.

$$\lim_{x\to\infty} 2^{\csc^{-1}(\ln x)}$$